

REVERSE ENGINEERING A MOUSE

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<https://github.com/dkbednarczyk/madr>

VXE MAD R

It's great!

- PAW3395 sensor
- up to 30,000 DPI
 - 8 stages
 - Independent X/Y values
- up to 8000hz polling rate
- Weighs 40g
- \$60



Its software sucks!

- Web driver
 - Online only, no PWA support
 - Some UI elements aren't translated even in the English version
- Desktop driver
 - Proprietary software from a Chinese company
 - Makes a network request every time you open it, even if you don't have an internet connection



will brown  

@willccbb



everyone's mice just went down



r/logitech · 2y ago
Playjasb2



Logi Options+ stuck loading forever

Support

UPDATE: To anyone who is getting this Jan 6th, 2026 issue, the solution I found here 2 years ago will not work. It's caused by some server issue...the fact that a server is absolutely needed for the mouse to have its functionality is quite baffling. It's embarrassing that they continue to have these long-standing issues while they continue to iterate on with their later mouse releases. It's not production ready. At this point, I'm considering using BetterMouse or other software, as it destroyed my Macbook Pro's battery life. Or if it comes to it, I'll switch to another mouse brand.

Methodology

1. Capture USB traffic with Wireshark
2. Keep changing one setting in the web configurator
3. Find the patterns in the bytes that change
4. Form a hypothesis, then test it
5. Repeat until the protocol is legible

HID feature reports are the way to communicate non-standard commands to a USB device. In this case we use them to set the configuration of the mouse.

Here's what the feature report looks like for setting the value of a DPI stage:

Byte	Value	Purpose
0-1	0x08 0x07	Header
2-3	0x00 0x00	Padding
4	variable	Identifies stage pair
5	0x08	Length of following stage data
6-9	variable	First stage data
10-13	variable	Second stage data
14-15	0x00 0x00	Padding
16	0x94 - report_id	Checksum

- DPI values between 100 and 30,000
- Increments of 50 DPI

One limitation of this is the maximum u8 value of 0xFF, (translating to 12,800 DPI).

Byte	Content
0	X DPI low byte
1	Y DPI low byte
2	High byte container
3	Checksum

Let's say we want to store 800 DPI for X and 15,000 DPI for Y.

The mouse doesn't store raw DPI values. Each unit in the protocol represents one 50 DPI increment above 100.

```
// Encoding 800
// 800 / 50 - 1 = 15 = 00001111
let x_val = (x / 50).saturating_sub(1);

// Encoding 15,000
// 15000 / 50 - 1 = 299 = 00000001_00101011
let y_val = (y / 50).saturating_sub(1);
```


The encoded value might need two bytes to store it, so it gets split into a low byte and a high byte.

Values under 256 (max u8) fit in one byte, so the high byte is always zero.

```
// x_val  00000000_00001111
let x_low = (x_val & 0xFF) as u8; // 00001111
let x_high = ((x_val >> 8) & 0xFF) as u8; // 00000000

// y_val  00000001_00101011
let y_low = (y_val & 0xFF) as u8; // 00101011
let y_high = ((y_val >> 8) & 0xFF) as u8; // 00000001
```

Both high bytes are packed into a single byte using bit shifts. X occupies bits 6–7 and Y occupies bits 2–3, leaving the remaining bits as padding.

```
/*                vv
x_high << 6      00000000
                ^^
```

```
                vv
y_high << 2      00000100
                ^^
```

```
high_container  00000100
*/
```

```
let high_container = (x_high << 6) | (y_high << 2);
```

The checksum is a wrapping subtraction of all payload bytes from the magic constant 0x55 (maybe chosen because of its nice binary representation of 01010101).

```
// 0x55 - x_low - y_low - high_container
// 0x55 - 0x0F - 0x2B - 0x04          = 0x17
let checksum = 0x55u8
    .wrapping_sub(x_low)
    .wrapping_sub(y_low)
    .wrapping_sub(high_container);
```

All we need to do now is send these values to the mouse in a feature report and we've set our new DPI value!

The mouse uses the same report structure for other settings as well:

- Polling rate
- Changing DPI stage
- Changing RGB settings

Any questions?

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